CHAPTER 3 – The Simplex Method

In LP the optimum solution is always associated with a corner point of the feasible region. Simplex method looks at the corner points (extreme points) in a systematic manner. It stops when no further improvement is possible.

Simplex method is best explained if the LP is in equation (standard) form.

In Equation (standard) form:

i. All constraints are equations with non-negative RHS. ✓
ii. All variables are non-negative. ✓

- To convert (≤) into (=) add a non-negative slack variable (s) to LHS;

$6x_1 + 4x_2 \leq 24 \Rightarrow 6x_1 + 4x_2 + s_1 = 24, \quad s_1 \geq 0$

If M_1 usage is less than 24, $s_1 > 0$, otherwise $s_1 = 0$

- To convert (≥) into (=) subtract a non-negative surplus variable (e);

$x_1 + x_2 \geq 800 \Rightarrow x_1 + x_2 - e_1 = 800, \quad e_1 \geq 0$

- If RHS is negative, multiply both sides by (-1);

$-x_1 + x_2 \leq -3 \Rightarrow -x_1 + x_2 + s_2 = -3, \quad s_2 \geq 0$

$\Rightarrow x_1 - x_2 - s_2 = 3, \quad s_2 \geq 0$

- Unrestricted variables

$x_1 + 2x_2 + x_3 = 10$

$x_1, x_2 \geq 0$

x_3 unrestricted

$\Leftrightarrow$

$x_1 + 2x_2 + (x_3^+ - x_3^-) = 10$

$x_3 = x_3^+ - x_3^-$

$x_1, x_2, x_3^+, x_3^- \geq 0$

- LP solution can never yield positive values for x_3^+ *and* x_3^- simultaneously
- If x_3 is in the objective function or other constraints it should be replaced by $(x_3^+ - x_3^-)$.

Transition from Graphical to Algebraic Solution

Graphical Method	Algebraic Method
Graph all constraints, including nonnegativity restrictions	Represent the solution space by m equations in n variables and restrict all variables to nonnegative values, $m < n$
Solution space consists of infinity of **feasible** points	The system has infinity of **feasible** solutions
↓	↓
Identify **feasible corner points** of the solution space	Determine the **feasible basic solutions** of the equations
Candidates for the optimum solution are given by a *finite* number of **corner points**	Candidates for the optimum solution are given by a *finite* number of **basic feasible solutions**
↓	↓
Use the objective function to determine the **optimum corner point** from among all the candidates	Use the objective function to determine the **optimum basic feasible solution** from among all the candidates

In the algebraic solution space (defined by m x n equations, m<n), **basic solutions** correspond to the corner points in the graphical solution space.
How can I extract the corner points algebraically?
They are determined by setting n-m variables equal to zero and solving m equations for the remaining m variables.

Corner points ←→ basic feasible solutions

m equations
n variables

In LP usually m<n
o/w if m=n → unique solution
m>n → at least m-n equations must be redundant

m<n

Set (n-m) variables to zero then solve the $m \times m$ equation system.

- Such a solution corresponds to a corner point.

Example:

Max $z = 2x_1 + 3x_2$

s.t.

$2x_1 + x_2 \le 4$

$x_1 + 2x_2 \le 5$

$x_1, x_2 \ge 0$

→

max. $z = 2x_1 + 3x_2$

s.t.

$2x_1 + x_2 + s_1 = 4$

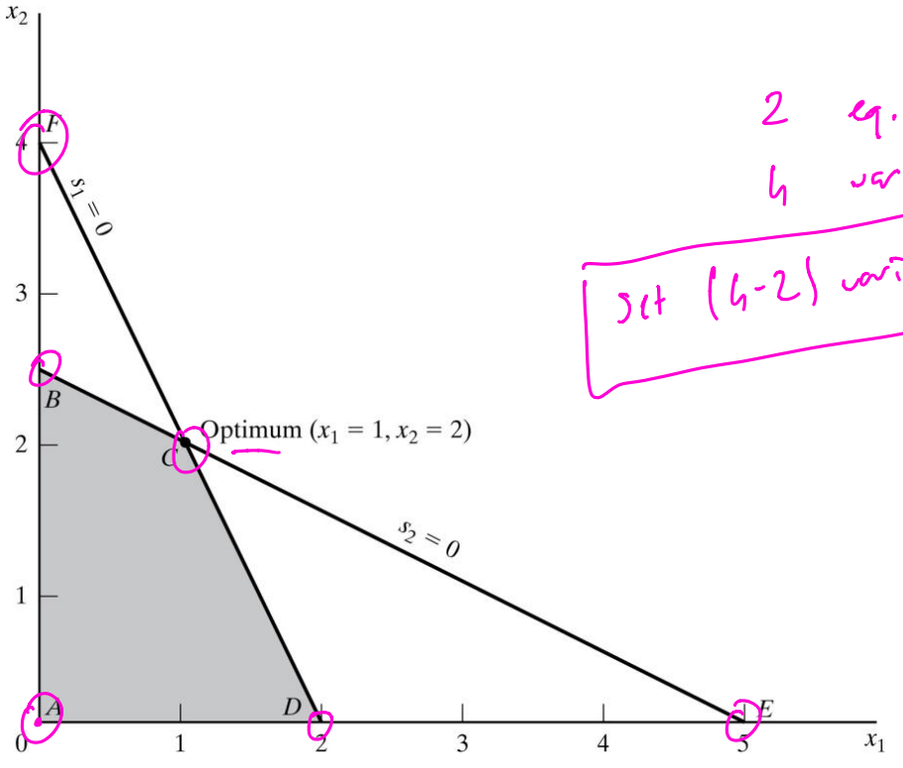
$x_1 + 2x_2 + s_2 = 5$

$x_1, x_2, s_1, s_2 \ge 0$

2 eq.

4 var.

Set (4-2) variables to zero



$$\max\ z = 2x_1 + 3x_2$$

s.t.

$$2x_1 + x_2 + s_1 = 4$$
$$x_1 + 2x_2 + s_2 = 5$$
$$x_1, x_2, s_1, s_2 \geq 0$$

set for example $x_1 = x_2 = 0 \rightarrow s_1=4,\ s_2=5$ } point A

Let us obtain the other corner points:

If $s_1=0=s_2$ then solve for $\left.\begin{matrix} 2x_1 + x_2 = 4 \\ x_1 + 2x_2 = 5 \end{matrix}\right\} \Rightarrow x_1=1,\ x_2=2 \rightarrow$ point C

			Feasible?	objective value
$x_1 = x_2 = 0$	$\Rightarrow$	$s_1=4,\ s_2=5 \rightarrow$ A	Yes	0
$x_1 = s_1 = 0$	$\rightarrow$	$x_2=4,\ s_2=-3 \rightarrow$ F	No	–
$x_1 = s_2 = 0$	$\rightarrow$	$x_2=2.5\ \ s_1=1.5 \rightarrow$ B	Yes	7.5
$x_2 = s_1 = 0$	$\rightarrow$	$x_1=2,\ s_2=3 \rightarrow$ D	Yes	4
$x_2 = s_2 = 0$	$\rightarrow$	$x_1=5,\ s_1=-6 \rightarrow$ E	No	–
$s_1 = s_2 = 0$	$\rightarrow$	$x_1=1,\ x_2=2 \rightarrow$ C	Yes	8

opt.

Note that E&F are infeasible!!!

- How many combinations are there for *m* equations and *n* variables?
- Which (n-m) variables should be set to zero?

(n-m) variables set to zero → non-basic variables
use m variables → basic variables → basic solution

If we are to investigate all basic solutions, it could be a computational burden.

$c\binom{20}{10} = 184{,}756$ sets of 10x10 equation systems should be solved.

20x10 is practically a small LP.

- Simplex method performs an intelligent search procedure among the extreme points.